

ESE 218 – Project 1

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I. INTRODUCTION

The goal of this project was to analyze electrical networks arranged in grid form by applying linear algebraic techniques. Specifically, I tried to compute node voltages and currents in a 5×5 resistive grid, implement my own LU factorization to solve the system of equations, and explore network properties such as power dissipation and effective resistance. This highlights the connection between Kirchhoff's circuit laws and matrix equations, and shows how computational methods can be applied to analyze large circuits.

II. BACKGROUND

Resistive networks can be explained by Kirchhoff's laws:

- **Kirchhoff's Current Law (KCL):** The total current flowing into a circuit junction, or node, must equal the total current flowing out of it.
- **Kirchhoff's Voltage Law (KVL):** The algebraic sum of all voltage changes (rises and drops) around any closed loop in a circuit is zero.

Applying KCL to each node (except those with fixed voltages) produces a system of linear equations with the form

$$Ax = b,$$

where A is the conductance (Laplacian) matrix of the network, x is the vector of node voltages, and b contains boundary conditions from fixed voltages. Each resistor between nodes i and j with resistance R_{ij} contributes a conductance $g_{ij} = 1/R_{ij}$ to A .

The Laplacian has a few important properties: it is symmetric, positive semi-definite, and each row sums to zero. Because of the last property, the system $Lx = 0$ is singular unless boundary conditions are specified.

To make the system useful, Dirichlet boundary is applied conditions at fixed-voltage nodes. If node k is fixed to voltage V_k^{fixed} , its row in the system is replaced by

$$A_{k,:} = \mathbf{0}, \quad A_{kk} = 1, \quad b_k = V_k^{\text{fixed}}.$$

This enforces $x_k = V_k^{\text{fixed}}$ exactly, while leaving the rest of the system unchanged. The resulting Dirichlet problem is nonsingular and has a unique solution for the voltages at all nodes.

III. METHODS

A. Formulating the System of Equations

The nodes are numbered $i = 1, \dots, 25$ row-by-row on the 5×5 grid, so that row r , column c corresponds to $i = (r - 1)5 + c$. If nodes i and j are connected by a resistor R_{ij} , the conductance is defined as

$$g_{ij} = \frac{1}{R_{ij}}.$$

Kirchhoff's Current Law states that the sum of currents leaving a node is zero:

$$\sum_{k=1}^r I_{ijk} = 0, \quad I_{ij} = g_{ij}(V_i - V_j).$$

Substituting Ohm's law yields

$$\sum_{k=1}^r g_{ijk}(V_i - V_j)_k = 0 \implies \sum_{k=1}^r g_{ijk}V_i - \sum_{k=1}^r g_{ijk}V_{jk} = 0,$$

which gives the coefficients of row i of the matrix A for any free node.

Each resistor (i, j) with conductance g_{ij} updates four entries of A :

$$A_{ii} += g_{ij}, \quad A_{jj} += g_{ij}, \quad A_{ij} -= g_{ij}, \quad A_{ji} -= g_{ij}.$$

This is the Laplacian effect on a resistor, making sure there is symmetry and current conservation.

If node k is fixed at voltage V_k^{fixed} , its KCL equation is replaced by the Dirichlet condition described previously.

Putting the rules together, the complete system is

$$A_{ij} = \begin{cases} 1, & i = j \text{ and } i \text{ is fixed,} \\ 0, & i \text{ is fixed and } j \neq i, \\ \sum_{k: (i,k) \in E} g_{ik}, & i = j \text{ and } i \text{ is free,} \\ -g_{ij}, & (i, j) \in E \text{ and } i \text{ is free,} \\ 0, & \text{otherwise,} \end{cases}$$

$$b_i = \begin{cases} V_i^{\text{fixed}}, & i \text{ fixed,} \\ 0, & \text{otherwise.} \end{cases}$$

letting E be the set of all given resistors.

In practice, the steps to formulate the system is:

- 1) Initializing $A = 0_{25 \times 25}$ and $b = 0_{25}$.
- 2) For each resistor (i, j, R_{ij}) , computing $g = 1/R_{ij}$ and applying the four updates above.

- 3) For each fixed node k , overwriting its row with a Dirichlet condition.

The resulting matrix A is symmetric and diagonally dominant, and after fixed rows are applied, the system $Ax = b$ is non-singular.

B. Reading files

The resistor file lists the circuit edges in the form

$$(i, j, R_{ij}),$$

meaning there is a resistor of value R_{ij} connecting node i and node j . This provides the edge set E and the resistance values used to compute conductance $g_{ij} = 1/R_{ij}$.

The fixed-voltage file lists pairs

$$(i, V_i^{\text{fixed}}),$$

meaning that at node i there is voltage V_i^{fixed} . This defines the set of Dirichlet boundary conditions that overwrite the corresponding KCL equations with $V_i = V_i^{\text{fixed}}$.

C. Assembling the A matrix.

Using the resistor data from the resistor file, the resistor rule from before is applied: for each edge (i, j) with resistance R_{ij} and conductance $g_{ij} = 1/R_{ij}$, update

$$A_{ii} += g_{ij}, \quad A_{jj} += g_{ij}, \quad A_{ij} -= g_{ij}, \quad A_{ji} -= g_{ij}.$$

Then, using the fixed-voltage set from the read voltage file, the corresponding rows of A are replaced with identity rows and set $b_i = V_i^{\text{fixed}}$. This produces the final linear system $Ax = b$.

D. LU factorization and triangular solve

Given a square matrix $A \in R^{n \times n}$, the LU factorization expresses A as the product

$$PA = LU,$$

where

- P is a permutation matrix encoding row swaps (used for partial pivoting),
- L is a unit lower triangular matrix ($L_{ii} = 1$ and $L_{ij} = 0$ for $j > i$),
- U is an upper-triangular matrix.

This decomposition is found by doing Gaussian elimination, which means eliminating entries below each pivot U_{kk} and recording the multipliers

$$L_{ik} = \frac{U_{ik}}{U_{kk}} \quad \text{for } i > k.$$

At each step, if the pivot U_{kk} is zero or very small, a row exchange is performed to maintain stability which is also recorded in matrix P .

Once the factorization $PA = LU$ is available, the system

$$Ax = b$$

is solved in two stages:

- 1) **Forward substitution:** First compute y by solving

$$Ly = Pb.$$

Since L is lower-triangular with ones on the diagonal, this is solved sequentially:

$$y_i = b'_i - \sum_{j=1}^{i-1} L_{ij}y_j, \quad \text{where } b' = Pb.$$

Each y_i depends on the previous computed entries.

- 2) **Backward substitution.** Then compute x by solving

$$Ux = y.$$

Since U is upper-triangular, this is solved in reverse order:

$$x_i = \frac{1}{U_{ii}} \left(y_i - \sum_{j=i+1}^n U_{ij}x_j \right).$$

Each x_i depends only on entries x_j with $j > i$, which are already known at that stage.

This procedure is similar to Gaussian elimination but stores the elimination steps in L and U so that the factorization can be reused. The combination of LU factorization with forward and backward substitution provides a systematic method to solve large systems $Ax = b$ arising from circuit equations.

E. Node voltages and link currents

Solving $Ax = b$ gives the vector x of node voltages. For each resistor (i, j, R_{ij}) , the current from i to j is computed using Ohm's law:

$$I_{ij} = \frac{V_i - V_j}{R_{ij}}.$$

F. Total power dissipation

The power dissipated in each resistor (i, j) is

$$P_{ij} = I_{ij}(V_i - V_j) = \frac{(V_i - V_j)^2}{R_{ij}}.$$

The total network power is

$$P = \sum_{(i,j) \in E} \frac{(V_i - V_j)^2}{R_{ij}}.$$

When all resistors have the same resistance R , this simplifies to $P = S/R$, where S is the voltage drop energy and depends only on the voltage distribution at $R = 1$, so plotting P versus R produces an inverse curve.

G. Scaling of incoming current

Boundary conditions $V_{\text{top-left}} = 1$ and $V_{\text{bottom-right}} = 0$ are imposed on a $k \times k$ grid. The incoming current at the bottom-right (BR) node is the sum of currents flowing into it from its neighbors ($\mathcal{N}$):

$$I_{\text{in}} = \sum_{j \in \mathcal{N}(\text{BR})} \frac{V_j - V_{\text{BR}}}{R}.$$

By solving grids of increasing size $k = 5, 10, 15, 20, 25$, it is observed how I_{in} scales with k .

H. Effective resistance between node pairs

The effective resistance between nodes a and b is defined as the voltage difference produced when a unit current is put in at a and taken out at b . It can be computed from the pseudoinverse L^+ of the Laplacian matrix L :

$$R_{\text{eff}}(a, b) = L_{aa}^+ + L_{bb}^+ - 2L_{ab}^+.$$

By applying this formula for all pairs (a, b) , the effective resistance matrix of the network is obtained.

I. Effective resistance scaling with R

If each edge has the same resistance R , then the effective resistance between two nodes scales linearly with R :

$$R_{\text{eff}}(a, b; R) = R \cdot R_{\text{eff}}(a, b; 1).$$

So, once the base effective resistance at $R = 1$ is computed the entire curve R_{eff} versus R can be plotted by simple scaling

IV. RESULTS

A. Node Voltages and Currents

After assembling A and b , x was solved for using the LU implementation. The results file contains the 25×25 system matrix A , the right-hand side b , the LU factorization components, node voltages, and currents through each link.

B. Power Dissipation

For a 5×5 grid with uniform resistance R , the total power dissipation follows $P(R) = S/R$ for some constant S determined at $R = 1$. Figure 1 shows the inverse relationship between P and R .

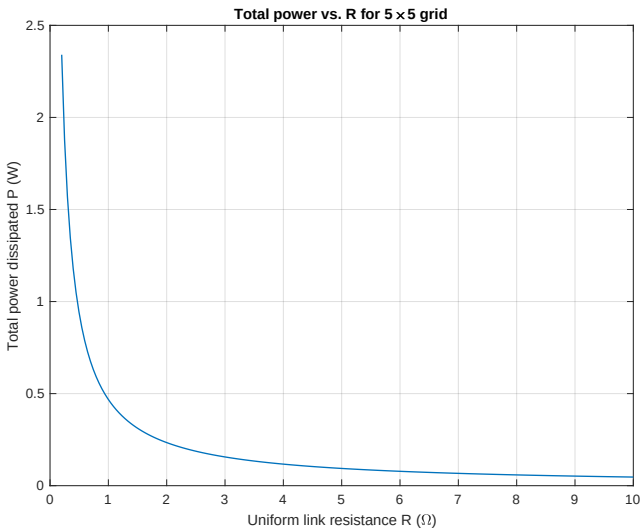


Fig. 1. Total power dissipated as a function of uniform link resistance R .

C. Scaling of Current with Grid Size

We fixed $V_{\text{top-left}} = 1$ and $V_{\text{bottom-right}} = 0$ and measured the incoming current to the bottom-right node. Figure 2 shows the increase in current as grid size k grows (for

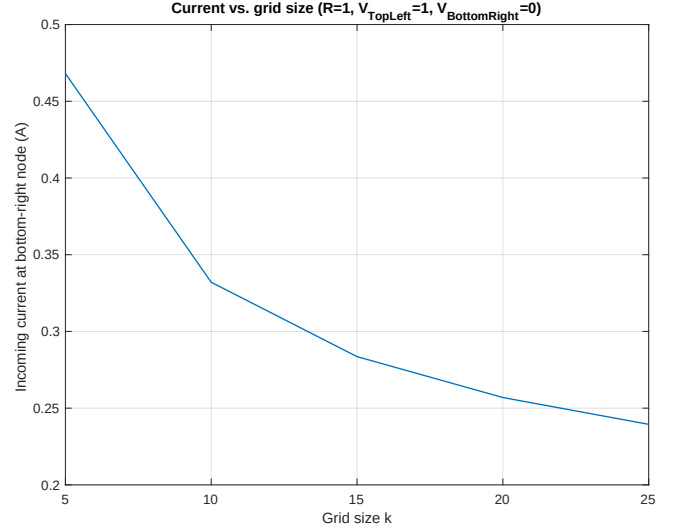


Fig. 2. Incoming current at the bottom-right node as a function of grid size k .

D. Effective Resistance

Using the pseudoinverse method, the effective resistance was computed between the top-left and bottom-right nodes for various grid sizes. Figure 3 shows the linear scaling of

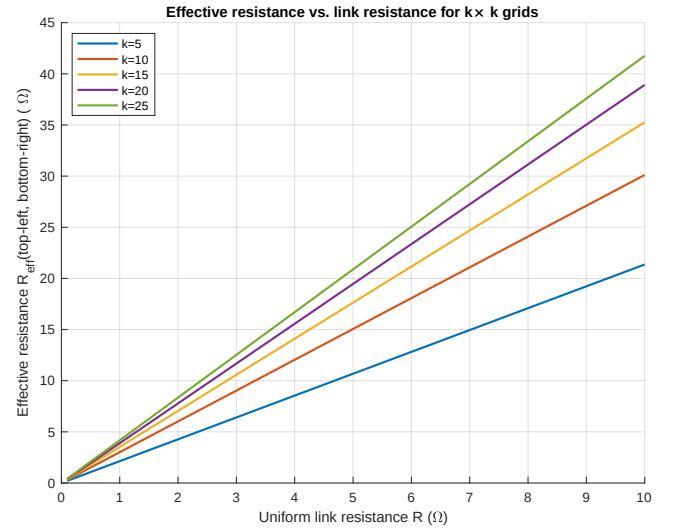


Fig. 3. Effective resistance between top-left and bottom-right nodes as a function of R .

V. CONCLUSIONS

This project demonstrated how Kirchhoff's laws naturally translate into a system of linear equations solvable by LU

factorization. I implemented custom solvers and applied them to a 5×5 grid, successfully obtaining node voltages and currents. It showed that:

- Total power dissipation scales as $1/R$ when all resistances are equal.
- The current into the bottom-right node increases with grid size, reflecting the greater number of parallel current paths.
- Effective resistance between opposite corners scales linearly with link resistance and decreases as the grid size increases.

These methods could extend to non-uniform or random resistor networks, analyzing sensitivity, and applying iterative solvers for very large grids.

VI. REFERENCES

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- [2] H. Li, “Properties and Applications of Graph Laplacians.” REU Paper, University of Chicago, 2022.